

Underwater hearing sensitivity of two ringed seals (*Pusa hispida*)

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Minimum audible field, underwater audiograms from 1 to 90 kHz were obtained for two ringed seals (*Pusa hispida*). The audiograms exhibited a uniform sensitivity, to within ± 7 dB, in the frequency range 1 to 45 kHz. Above 45 kHz the threshold increased at a rate of 60 dB per octave. The lowest threshold was -32 dB relative to 1 μ bar (68 dB re 1 μ Pa) at 16 kHz. The audiograms were similar to those of other phocid seals so far studied.

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On a déterminé le champ audible minimum et obtenu des audiogrammes sous l'eau de 1 à 90 kHz, chez deux phoques annelés (*Pusa hispida*). Les audiogrammes reflètent une sensibilité uniforme, de l'ordre de ± 7 dB, dans l'étendue des fréquences 1 à 45 kHz. Au-dessus de 45 kHz, le seuil augmente au rythme de 60 dB par octave. Le seuil minimum est de -32 dB re 1 μ bar (68 dB re 1 μ Pa) à 16 kHz. Les audiogrammes obtenus sont semblables à ceux d'autres phocidés étudiés précédemment. [Traduit par le journal]

Introduction

Underwater hearing thresholds of harbor (*Phoca vitulina*), harp (*Pagophilus groenlandicus*), and grey (*Halichoerus grypus*) seals exhibit great similarities (Møhl 1968; Terhune and Ronald 1972; Ridgway and Joyce 1974). The underwater hearing of the California sea lion (*Zalophus californianus*) is less sensitive at and above 32 kHz than is that of the three phocid species (Schusterman *et al.* 1972). It is possible that at the family or subfamily level the hearing abilities of pinnipeds are basically the same. Consistent differences likely occur between members of different subfamilies. An examination of the underwater sensitivity thresholds of the ringed seal (*Pusa hispida*) would indicate whether or not its hearing sensitivity is the same as the other members of its subfamily (Phocinae) so far studied. Two ringed seals were examined in the present study. This was done so that estimates of individual as well as interspecific differences could be made. Knowledge of the ringed seal's hearing ability may prove useful in assessing the influence of sounds on its behavior, such as the evasive actions resulting from noises made by either of its two main predators, polar bears (*Thalarctos maritimus*) or man.

Materials and Methods

Two healthy 3-year-old ringed seals (a male and a female) were the subjects of minimum audible field

underwater audiogram determinations. These were obtained by a method of yes-no response to variable stimuli using methods described by Møhl (1968) and Terhune and Ronald (1972) and a statistical analysis described by Wetherill and Levitt (1965).

The seals were housed in an indoor, plastic resin-coated, wooden tank $4 \times 3 \times 1.2$ m deep. A nylon net separated the tank into two equal-sized areas (2×3 m). One seal and a set of three switches (see later) were located in each area.

Underwater sounds were generated by using a broad-bandwidth, omnidirectional hydrophone (Atlantic Research LC-32) as a sound source. This hydrophone was centrally supported at a depth of 0.5 m by the nylon net. The hydrophone was driven by a Wavetek 112 function generator (producing a continuous sine wave) through a photoresistor gate, followed in turn by a precision attenuator and a filter (Krohn-hite 3550). The filter was included to reduce noise and harmonic levels. Control of the photoresistor gate was exercised by a seal pressing its nose against a "stimulus" switch. This switch was 0.5 m from the sound source, 0.5 m under the water surface, and at least 1.5 m from the nearest wall of the tank. The attenuator controlled the intensity of the sound that was generated while the "stimulus" switch was closed. The attenuator decreased the sound levels by discrete increments as follows: from 0 dB to -8 dB, then from -8 dB to -30 dB in 2-dB steps; from -31.5 dB to -41 dB in either 1.5- or 2-dB steps, then finally to -50 dB. The maximum attenuation setting of 50 dB was used for the "no sound" or "catch trial" part of the experiment described later.

The sound pressure levels and wave forms of the signals were determined by placing a calibrated hydrophone (Atlantic Research LC-32) at the place where the seal's head would be while it pressed the "stimulus" switch. The signal received was amplified by a pre-amplifier (Tektronix FM122) and passed through a

band-pass filter (Dawe 1471) to a voltmeter (Bruel and Kjaer 2409) and an oscilloscope (Tektronix 510 3N).

Each seal was trained to push the "stimulus" switch voluntarily. This brought the seal's head into a relatively standard position for each trial. As long as this switch was depressed, the photoresistor gate was open and the preset signal was passed through the attenuator and filter and was broadcast by the hydrophone. If the seal detected a signal it was trained to push a response switch ("yes") located some distance away. If the seal did not detect a signal (catch trial or signal error) it pushed a second response switch ("no") located near the "yes" switch. The seal was rewarded with a piece of fish after each correct response. If the response was incorrect, the reward was withheld and a lamp, visible to the seal, lit. The seals received food only during training or testing sessions.

The initial training was considered complete when the seal correctly responded to both clearly audible signals and catch trials at least 80% of the time. After this, each seal was pseudotested for five sessions (see below) to familiarize it with the final threshold determination task. The data from these five sessions were not used to calculate the thresholds.

The thresholds (sound pressure levels detected 50% of the time) were estimated by an adaptive psychophysical procedure (Wetherill and Levitt 1965). After each correct response to a signal the sound pressure level of the next signal to be presented was automatically reduced by a 2- or 1.5-dB step. After each incorrect response to a signal, the sound pressure level of the next signal to be presented was automatically increased by a 2- or 1.5-dB step. Correct or incorrect responses to catch trials did not alter the signal levels. The seal was first presented with sound pressure levels well above threshold. The signal levels were decreased until the first signal error was made. Subsequently, the signal levels were increased until the seal responded correctly. The first correct signal response following the first signal error was counted as the start of the first run. A run was defined as a series of attenuator increments in one direction, e.g. from the level of the first correct signal (peak or highest sound level presented) after one or more signal errors to the level of the next signal error (valley or lowest sound level presented). Each session contained 10 consecutive runs. The estimate of the threshold was obtained by averaging the peaks and valleys of the 10 runs. The variation was estimated by the standard deviation from the mean of the peaks and valleys (Wetherill and Levitt 1965).

The sound pressure levels were set by the experimenter's "best guess" prediction based on previous experience (Terhune and Ronald 1972) so that the seal's threshold would be near the -20-dB position of the attenuator. In this way the number of trials before the first signal error were not too numerous and most of the signals were present in the -2-dB step range of the attenuator.

The seal had an equal chance of being presented with a signal or a catch trial. The order of the presentations was predetermined and random within the limit of four consecutive signals or catch trials. The presentations were automatically controlled by a punch-tape reader circuit. Each seal was tested once per day (the female

first for behavioral reasons). Each test of 10 runs required 75-100 trials. The underwater hearing sensitivity was determined for each seal for frequencies between 1 and 90 kHz as per Table 1. The upper and lower frequency limits were imposed by the apparatus and not by the seals.

Results

At all frequencies, standing waves and echoes caused a 5- to 15-dB variation in the sound field.

The thresholds are given in Table 1. Generally the behavior of both seals appeared quite consistent. The seals would occasionally push the "stimulus" switch twice before going to either of the response switches. The number of correct catch trials varied greatly (Table 1). A few times the seals, regardless of the stimulus, responded only to the "yes" or "no" switches or stopped pressing the switches altogether. The data obtained during these sessions were discarded.

In two-thirds of the sessions the peaks and valleys of the runs were all within the -2 dB per step range of the attenuator. While testing the female at 90 kHz, 8 of the 10 peaks were at the 0-dB attenuator level, hence the first step was -8 dB and the other steps were -2 dB.

Discussion

The sound-level calibrations were based on the highest sound pressure level consistently recorded in the sound field. It was assumed that the seal would locate the area of maximum intensity within the fluctuating sound field (Johnson 1967).

Throughout the testing period an effort was made not to preferentially influence the responses of the seal when it was presented with catch trials. That is, the seal was not punished (by stopping the session early, for example) if it made a high number of catch trial errors. This was done so the seals would not be encouraged to establish a criterion which would bias their responses toward a catch trial response. Such a situation (termed the "Newman-Pearson animal") is believed to have occurred in many marine mammal psychophysical threshold determinations and may have resulted in underestimating the subject's threshold (Schusterman 1974). Judging by the percentage-correct catch trials (Table 1), the seals altered their response criteria from one session to the next. It is possible that in some instances the

TABLE 1

Underwater hearing sensitivity of two ringed seals in dB re 1 μ bar. To convert to dB re 1 μ Pa, add 100

Frequency, kHz	Threshold, dB re 1 μ bar		SD, dB		Catch trials, % correct	
	♀	♂	♀	♂	♀	♂
1.0	-26	-24	3.1	3.1	53	68
1.4	-22	-20	4.0	2.7	61	80
2.0	-19	-22	3.0	3.0	64	73
2.8	-20	-19	3.5	3.1	73	91
4.0	-25	-25	2.9	2.6	80	86
5.6	-23	-28	2.2	4.8	86	77
8.0	-20	-26	2.5	2.8	77	78
11.3	-32	-28	6.2	7.6	44	50
16.0	-32	-28	7.8	5.7	68	49
22.9	-27	-29	3.2	7.0	76	67
32.0	-21	-29	2.8	2.6	87	83
44.9	-25	-31	4.8	2.9	55	85
55.0	-11	-14	4.9	4.1	69	100
64.0	+15	+4	3.7	3.6	58	80
90.0	(+18)	+12	(5.1)	3.0	61	66

seals were acting as "Newman-Pearson animals." It is not possible to determine the influence this might have had on the threshold estimates. For a few of the determinations the responses of either of the seals to the catch trials were at, or close to, chance levels. In these instances the seal's responses to the various signal levels were not (subjectively) abnormal. Because of this and the fact that the threshold estimates and their standard deviations were similar in magnitude to those determined for adjacent frequencies (for which the number of correct responses to catch trials were well above chance levels) it was assumed that all of the determinations would be considered valid. It was thought that if certain results were disregarded, a non-uniform bias would be imposed which would be equivalent to creating a "Newman-Pearson animal" situation. The ringed seal thresholds were similar to those of the grey seal (which would not be influenced by response criteria because of the methodology), indicating that the influence of the seal's response criteria on the threshold estimates was not of major importance.

The dynamic range of the testing system was 33 dB (i.e. attenuation settings between 8 and 41 dB). The difficulty encountered at 90 kHz was that the maximum sound pressure level that could be produced was barely above

the female seal's threshold. Had the sound pressure level been greater or the seal more sensitive, then the frequency could have been estimated using only 2-dB steps. The relatively wide testing range provided by the attenuator meant that the seals could, if they were more sensitive than the experimenter's prediction, be tested at lower levels without interrupting the session to adjust the sound pressure levels. This was not possible in a similar experiment with a harp seal and a testing range of only 14 dB (Terhune and Ronald 1972). If the seal's responses went beyond the range of the attenuator the threshold determination could not be made. This not only resulted in a loss of data but it, in effect, restricted the variability of the seal to 14 dB from the highest peak to the lowest valley. In the present experiment, the sound intensities presented and the responses of the seals were such that the subjects never reached the lowest sound pressure level that was available. That is, the attenuator setting of 41 dB was never used. This indicated that the sound pressure levels set by the experimenter were not limiting the responses by the seal. This may account for the differences in the size of the standard deviations of the threshold estimations between this study and those of the harp seal study of Terhune and Ronald (1972).

Despite methodological differences, the pres-

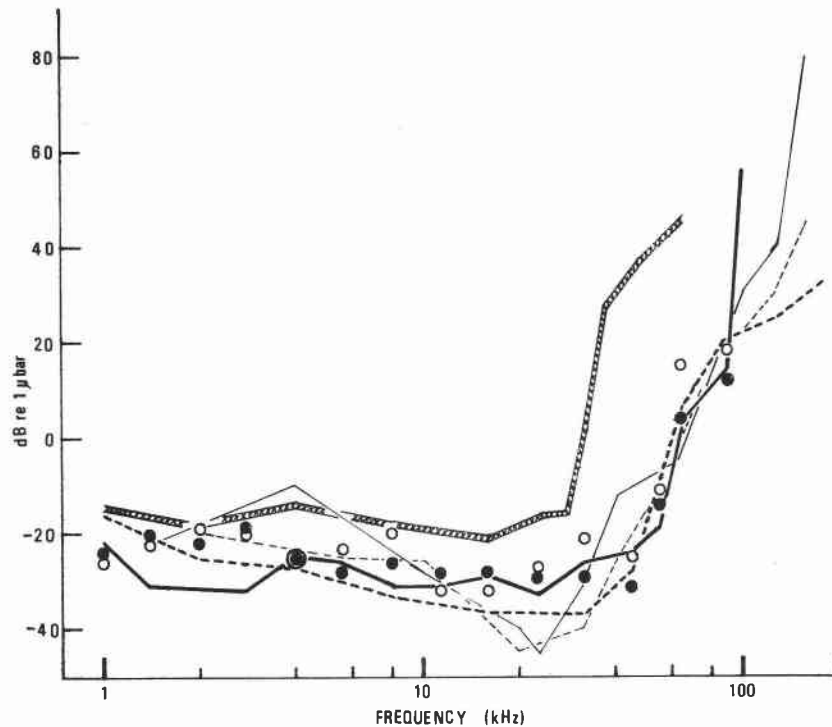


FIG. 1. Underwater audiograms of the female (○) and male (●) ringed seals; a harp seal (—), a harbor seal (---), two grey seals (—) and (---), and a California sea lion (×).

ence of standing waves in the sound field, and variations in individual animals, the underwater hearing sensitivities of the two ringed seals, the harbor seal (Møhl 1968), the harp seal (Terhune and Ronald 1972), and the two grey seals (Ridgway and Joyce 1974) were similar. This is especially true in the 45- to 90-kHz frequency band (Fig. 1). The least similarity occurred below 8 kHz and above 100 kHz. The presence of standing waves in the test area (especially at the lower frequencies where the variation in the sound field was usually greatest) and (or) the presence of a tuning or resonance phenomenon (Møhl 1968; Terhune and Ronald 1972) may account for much of this variability. From 1 to 90 kHz, however, the four phocid species do not exhibit differences in sensitivity (at any particular frequency) greater than 20 dB. Fletcher (1940) reports that 50% of a group of humans have a hearing sensitivity in air that is within ± 10 dB of the average of the group. Assuming that seals exhibit a similar range of variation, the variability among the four phocid species could be attributed to individual

and experimental differences. The audiogram of the California sea lion (Fig. 1) is different in that there is much less sensitivity at the higher frequencies. This may be a result of this species having an upper frequency discrimination limit at a lower frequency (Schusterman *et al.* 1972). It may therefore be possible to talk in terms of "phocid" vs. "otariid" hearing (underwater at least) and assume that all members of the same family or subfamily will have virtually identical hearing abilities. Further experimentation will be required before this assumption can be confidently applied to hearing aspects other than underwater sensitivity.

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